

Shangmin Chen

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EDUCATION

Boston University

Bachelor of Arts in Computer Science

Boston, MA

September 2022 – May 2026

EXPERIENCE

Software Engineer

BU Spark!

September 2024 – December 2024

Boston, MA

- Developed full-stack food waste platform using Django, JavaScript, and SQLite, connecting 500+ students and reducing campus waste by 10%
- Built RESTful API with sub-100ms query times and deployed via CI/CD pipelines on Heroku
- Implemented Auth0 authentication and Django security middleware, ensuring zero security incidents
- Accelerated release by 1 week using Agile methodology in team of 5, contributing to 7 iterations

IT Consultant

Boston University Engineering IT

January 2024 – July 2025

Boston, MA

- Automated software deployment for 100+ devices using Microsoft Deployment Toolkit and created custom USB deployment solution for Windows 11, reducing deployment time by 40%
- Managed 200+ user accounts and permissions via NAS, conducting access audits and boosting system uptime by 15%
- Enhanced endpoint security with CrowdStrike for 50+ devices, resolving 20+ tickets/week with 100% incident resolution rate

PROJECTS

High-Frequency Market Data Processor | *C++, Boost Libraries, Custom Memory Pool, Lock-Free Queues*[Month] 2026

- Engineered low-latency market data processing engine with custom memory allocators and lock-free data structures, achieving [X] ns average processing time per message (benchmark against system clock)
- Implemented multi-threaded order book reconstruction with NUMA-aware thread pinning and CPU cache optimization, supporting [X]K messages/second throughput (stress test with synthetic data generator)
- Built real-time technical indicator calculations (SMA, EMA, VWAP) using SIMD instructions and template metaprogramming, reducing computation overhead by [X]% (compare with naive implementation)

Scalable Chat Platform | *Go, AWS (ECS, RDS, ElastiCache, ALB), WebSockets, Docker* [Month] 2026

- Architected cloud-native messaging platform on AWS ECS with auto-scaling policies, PostgreSQL RDS, and Redis ElastiCache, handling [X] concurrent WebSocket connections (load test with Artillery.js)
- Implemented horizontal scaling with Application Load Balancer and sticky sessions, deploying via GitHub Actions CI/CD with blue-green deployment strategy achieving [X]% uptime (CloudWatch metrics)
- Integrated comprehensive observability with CloudWatch dashboards, custom metrics, and structured logging, enabling sub-[X]ms P95 response time monitoring (APM traces and database query analysis)

Real-Time Analytics Pipeline | *Apache Kafka, Apache Flink, InfluxDB, React Dashboard, Docker Compose*[Month] 2026

- Built event-driven analytics system processing [X]K events/minute through Kafka topics with Flink stream processing, implementing windowed aggregations and complex event pattern detection (producer load testing)
- Designed time-series data storage in InfluxDB with optimized retention policies and continuous queries, supporting [X]-day historical analysis with [X]ms query response times (benchmark with InfluxDB CLI)
- Created real-time React dashboard with WebSocket streaming and D3.js visualizations, displaying live metrics with [X]ms end-to-end latency from event ingestion to UI update (browser performance profiling)

Audio Transcription Platform | *Java Spring Boot, Python FastAPI, React, PostgreSQL*

September 2025

- Built microservices platform with Spring Boot API, Python FastAPI service, and React frontend, implementing async audio processing with OpenAI Whisper and real-time status polling
- Designed PostgreSQL database with 4 related tables, Flyway migrations, and optimized indexing, supporting multiple audio formats with comprehensive error handling

- Deployed containerized application using Docker Compose with inter-service communication, CORS configuration, and health monitoring endpoints

Fitness Tracking Mobile App | *React Native, TypeScript, Expo*

August 2025

- Built cross-platform app with drag-and-drop workout creation using React Native Reanimated v3 and PanGestureHandler for exercise reordering
- Implemented persistent data storage with AsyncStorage and custom React hooks for state management across workout sessions
- Developed modular component architecture with search/filter functionality and modal-based exercise selection interface

Crime Analytics & Forecasting Platform | *Python, Streamlit, Prophet, Scikit-learn, GeoPandas* December 2024

- Built crime analytics platform processing 500K+ Boston crime records with Prophet time-series forecasting and DBSCAN clustering for hotspot identification across 12+ police districts
- Implemented advanced data pipeline with seasonal decomposition and confidence intervals, generating 2-month crime predictions with train/test validation
- Developed interactive dashboard using Streamlit and Plotly, creating 121 crime-type heat maps with Folium and real-time geospatial visualizations

Decentralized ML Model Marketplace | *Solidity, Circom, IPFS, Express.js, TensorFlow*

May 2024

- Built decentralized marketplace using Solidity smart contracts with ERC-20 tokens, implementing 6-state transaction management, automated candidate selection, and economic incentive mechanisms with stake slashing
- Integrated IPFS distributed storage with Ethereum blockchain using Thirdweb SDK, enabling secure model uploads with SHA256 hash verification and automated ML training workflows
- Implemented zero-knowledge proof system using Circom and Groth16 protocol for privacy-preserving accuracy verification, with MetaMask integration and real-time blockchain event monitoring

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL (PostgreSQL), JavaScript, TypeScript, HTML/CSS, Solidity

Frameworks & Libraries: React, React Native, Node.js, Spring Boot, FastAPI, Django, Streamlit, Prophet, Scikit-learn

Technologies & Tools: Docker, Git, PostgreSQL, SQLite, IPFS, Auth0, Heroku, OpenAI API, GeoPandas, Plotly

Platforms & Systems: AWS, Ethereum, MetaMask, CrowdStrike, Microsoft Deployment Toolkit, NAS